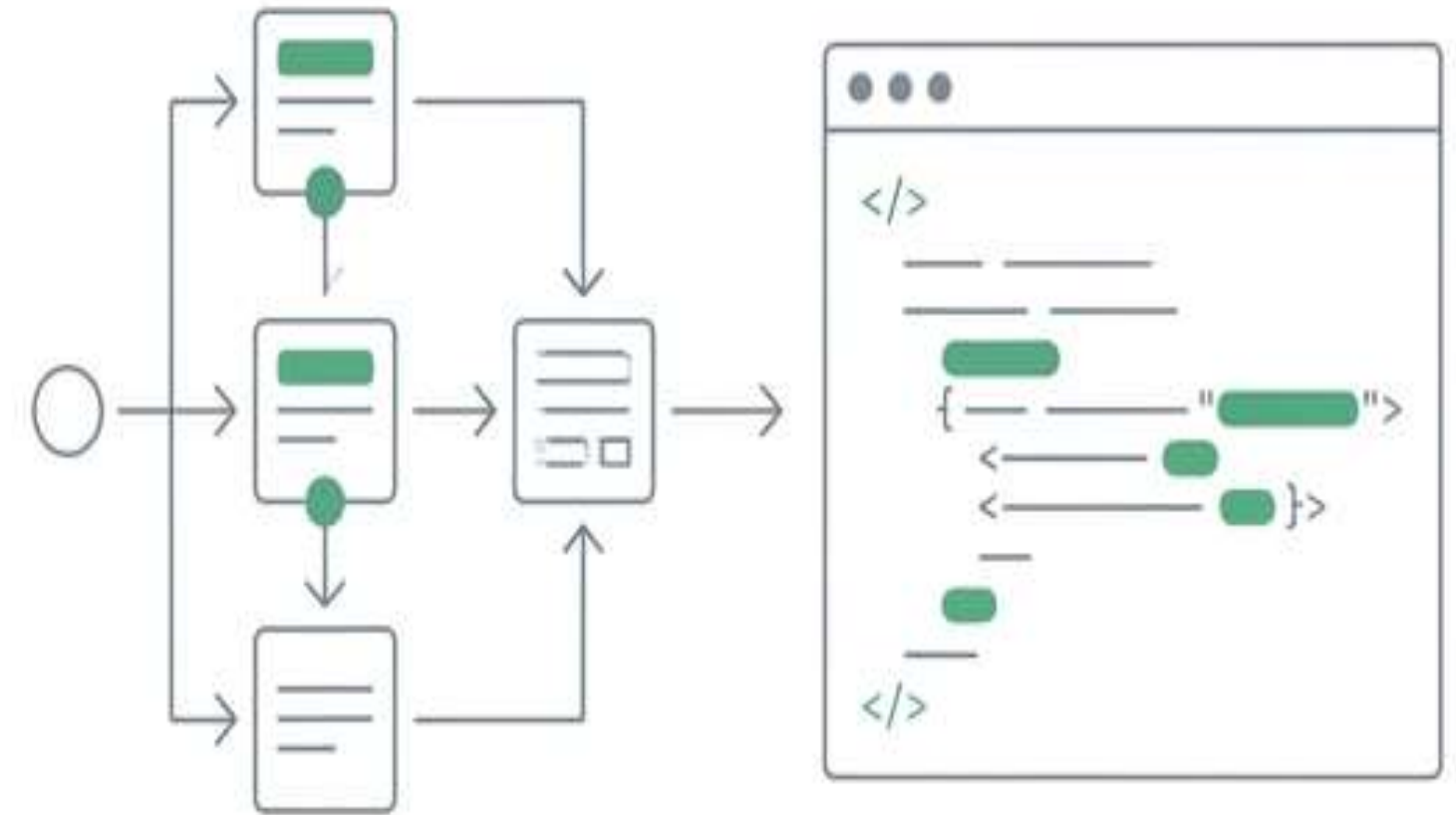
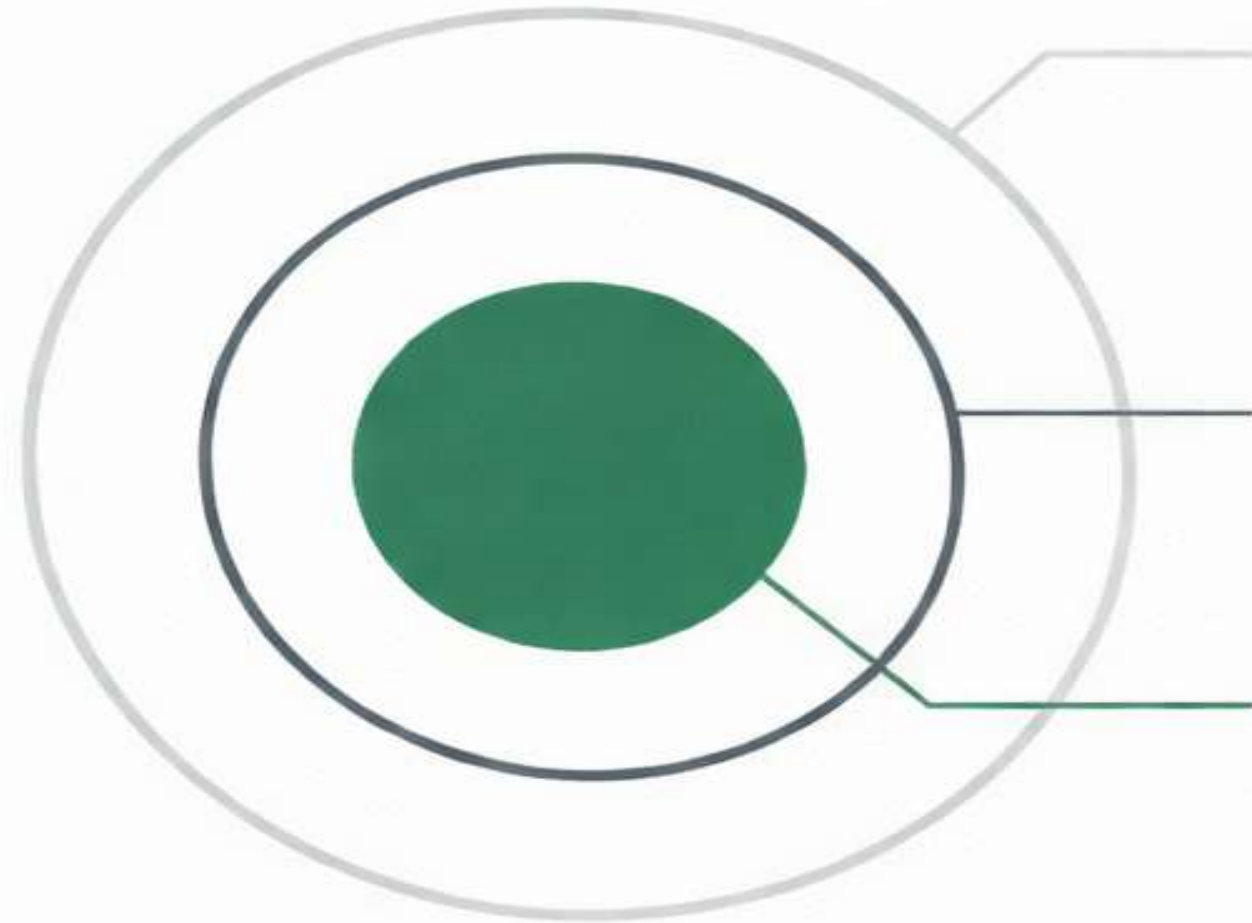


Advanced Data-Driven Strategies in Selenium WebDriver



The Case for Dynamic Parametrisation



What: Validate same functionality across multiple data sets.

How: TestNG Annotation Engine.

Why: Ensure comprehensive test coverage and reliability.

Efficiency

Execute logic repeatedly with varying inputs.

Coverage

Validate edge cases without code duplication.

Scalability

Manage config changes and datasets easily.

The Mechanics of @DataProvider

- **Definition:** A TestNG annotation that supplies data to test methods.
- **Data Structure:** Returns a 2-Dimensional Array (Object[][] or String[][]).
- **Role:** Acts as a feeder, injecting arguments into the @Test method.

Parameters (Input Arguments)

	Param 1	Param 2	Param 3	Param 4
1	User1	Pass1	RoleA	True
2	User2	Pass2	RoleB	False
3	User3	Pass3	RoleC	True

Iterations (Test Execution Count)

→ @Test (dataProvider = '...')

Anatomy of the Data Provider Method

```
@DataProvider(name='getData')  
public String[][] fetchData() {  
    String[][] data = new String[2][4];  
    data[0][0] = 'TestLeaf';  
    data[0][1] = 'Hari';  
    // ... data population  
    return data;  
}
```

The Annotation
(Essential)

Array Dimensions
(Rows x Cols)

Return Statement
(Feeds the Runner)

Implementation Workflow: 5-Step Process



Practical Scenario: CRM Lead Creation

The Data (Array)

	Company Name	First Name	Last Name	Phone Number
1	TestLeaf	Hari	R	99
2	TestLeaf	Babu	Manickam	91

The Application
(Web Form)

Create Lead

Company Name

[Input Company Name]

First Name

[Input First Name]

Last Name

[Input Last Name]

Phone Number

[Input Phone Number]

Submit

Professional Best Practices

✓ Naming Conventions

Use **descriptive names** like "fetchLoginData" instead of generic defaults.

✓ Type Safety

Prefer `Object[][]` over `String[][]` for mixed data types (int, boolean).

✓ Data Separation

Keep Data Providers in a separate "**DataUtils**" class for enterprise frameworks.

✓ Maintainability

Avoid hard-coding massive arrays; integrate Excel/JSON for large sets.

Interview Preparation: Key Concepts

Return Type?

Q: What does a DataProvider return?

A: A 2D array of Objects (**Object[]**) or **Strings**.

Execution?

Q: How to run same test with different data?

A: Discuss **@DataProvider** annotation vs **parameters** in **testng.xml**.

Mapping?

Q: How do parameters match?

A: Arguments in **@Test** must strictly match **columns in the array**.

Summary of Dynamic Parametrisation

